Lecture 1 (9/14)

Intro to Math 50 and Unit 1

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Office Hours: Tuesday 8–10am, 11am–12:30pm

Math 50: Regression models from stats to ML

- Regression modeling: Identifying explanatory variables from empirical data.

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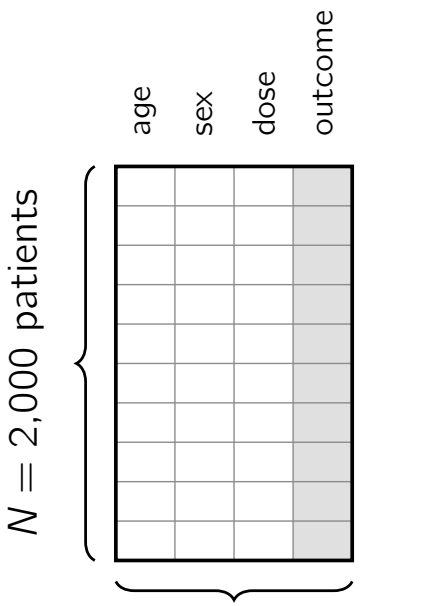
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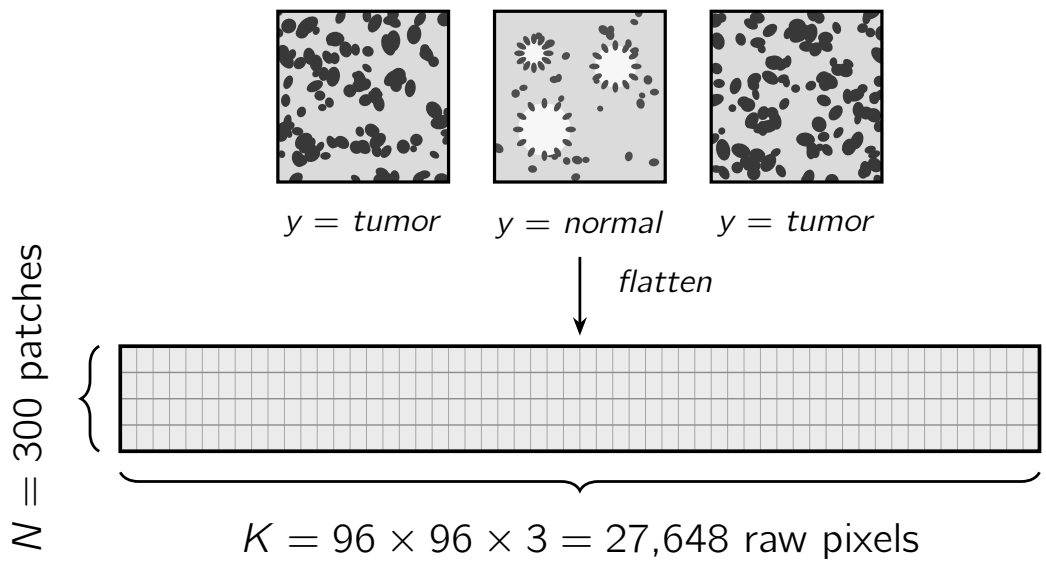
STATISTICS

clinical trial



MACHINE LEARNING

histopathology study



Statistics vs. Machine Learning

	Statistics	Machine Learning
Goal	Inference	Prediction
Data regime	$N > K$	$K > N$
Model	Linear	Nonlinear
Estimation	Formula	Gradient descent
Latent space	Low-dim	High-dim

Overlap with other courses at Dartmouth

- MATH 40: More mathematically detailed look at statistical inference (overlap with Units 1–2). More analytical calculations.
- MATH 70: Higher level, covering methods in statistics and machine learning (ML) such as classification and unsupervised learning (overlap with Units 4–5).
- ECON 20: More focused on causal inference (overlap with Unit 4).
- CSC 74: More broadly focused on ML, both supervised and unsupervised (overlap with Unit ≥ 5).

Workload/Grading

- **Weekly assignments (15%):** Longer multipart questions that may involve working with real data and thinking more conceptually.
- **Exams (75%):** Midterm 1 (20%), Midterm 2 (20%), Final (35%) — Lower midterm score is replaced by the final if final is higher.
- **Attendance and participation (10%):** checked periodically; lateness/absence can affect this grade.
- **Exercises (not graded):** drills and end-of-unit problems — not graded.

Course policies

- **No computers or phones in the classroom unless I explicitly ask for you to do something on the computer**. NO exceptions unless you have documentation.
- All course notes are available prior to class and can be printed out before the lecture, if you would like to follow notes in class.
- I will ask you to leave the room if you are using your phone or computer. I also may penalize you on the participation grade.

AI policy

- Free to use LLMs (ChatGPT, Claude, etc.) on take-home assignments — but unless stated otherwise, you should be able to complete them without one, and **all LLM output must be reviewed and edited**: padded, "tropey" answers can cost you points even if correct.
- If you use an LLM, include a short *disclosure statement*: which model, how you used it, and how you engaged with its output.
- No LLMs or electronics during in-class exams or lecture.
- Full policy and rationale on the course website.

Unit 1: Goals

- Distinguish a model from the world it represents
- Identify sample spaces, outcomes, events, and random variables for a given modeling situation.
- Read and write probability-distribution notation (using $\sim$ vs P_X).
- Understand how to use properties of probability distributions to calculate probabilities of events.

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- Probability or statistical models are mathematical models involving random variables (e.g. a fair coin is heads 1/2 of the time and tails 1/2 of the time)
- Statistics is about using data to determine what good probability models are for things in the real world.

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- Y takes values in a sample space S. The things in the sample space are called outcomes. Sets of outcomes are events. For an event E, we write $P(E)$ for the probability that E happened.
- A probability distribution P_Y assigns probabilities to outcomes for Y: For $y \in S$

$$P_Y(y) = P(\{Y = y\}), \quad P_Y(y) \geq 0, \quad \sum_y P_Y(y) = 1.$$

Probability distributions

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Question

Let Y be the roll of a fair die, so $S = \{1, 2, 3, 4, 5, 6\}$ and $P(Y = y) = 1/6$ for each $y \in S$. Some event probabilities:

- $P(Y \text{ even})$
- $P(Y > 4)$
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Question

- Two people are sampled from a population and are asked a Y/N question. Their response is recorded in binary variables X and Y. What is the sample space of (X, Y)?
- IF it is known that a fraction $q \in [0, 1]$ of people in the population from which they are sampled answer yes, what are all the probabilities of the outcomes?

Example

A basketball player with a career free-throw percentage of 75% shoots N free throws. The outcome of each shot (make or miss) is modeled as a Bernoulli random variable with probability of success (making the shot) $q = 0.75$.

Question

Let $C \in \{0, 1\}$ indicate cancer diagnosis among a cohort, and $G \in \{M, F\}$ denote gender. Data gives us

$P(C, G)$	$G = M$	$G = F$
$C = 1$ (diagnosed)	0.0021	0.0019
$C = 0$ (not diagnosed)	0.4899	0.5061